

Genetic Algorithm – Next Generation (7 marks)

Problem statement (brief)

A simple Genetic Algorithm (GA) uses 6-bit binary chromosomes. The objective function to maximize is:
Initial population (population size = 5):

ID	Chromosome	Fitness
1	110101	27
2	101110	26
3	011011	18
4	111100	30
5	010101	20

Crossover probability: $p_c = 0.8$

Mutation probability per bit: $p_m = 0.05$

Use roulette-wheel selection and one-point crossover to produce the next generation (5 offspring) and compute their fitness.

1) Selection by roulette wheel

Total fitness = $27+26+18+30+20 = 121$.

Cumulative probabilities: $C1=0.2231$, $C2=0.4380$, $C3=0.5876$, $C4=0.8347$, $C5=1.0000$.

2) Pairing and one-point crossover

Pair selected list sequentially: (1,3), (4,2), (5,1). Crossover occurs with probability 0.8.

Pair (1:110101) $\times$ (3:011011): cut=2 $\rightarrow$ Children: 111011, 010101.

3) Mutation

Mutation rate = 0.05 per bit. Expected flips = 1–2.

4) Fitness calculation

Using $f(x) = 2x_1 + 3x_2 + 4x_3 + 5x_4 + 6x_5 + 7x_6$:

Final Next Generation:

Offspring	Binary	Fitness
1	111011	22
2	010100	8
3	111100	14
4	101110	17
5	011101	19

Conclusion: Roulette wheel selection ensures fitter chromosomes are more likely to reproduce while maintaining diversity. The resulting offspring demonstrate the effect of crossover and mutation in evolving better solutions.